6 Solids, Liquids, and Gases

Everything present around us is matter. This may include the things that we can see (chair, table, glass, book, etc.) as well as the things we cannot see (like air around us). Matter is itself made up of lots of very tiny particles called molecules. Each molecule is further made of particles known as atoms. The states of these matter depend upon the arrangement of their constituent molecules.

Learn about

- States of matter
- Solids
- Liquids
- Gases
- Solution
- Soluble and insoluble substances
- Separation of liquids from solids
- Air
- Properties of air
- Air rises on heating

States of Matter

Solid, liquid, and gas are the three states of matter. We know that chair, book, and fan are solids. Water, ink, and oil are liquids. Oxygen, nitrogen, and carbon dioxide are gases. Let us now study in detail how these three states of matter are different from each other.

Fact File

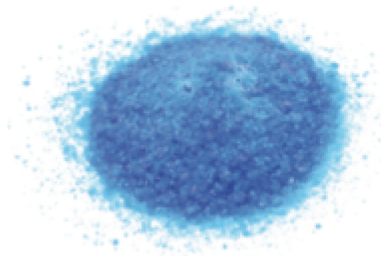
Do you know two more states have been discovered by scientists! **Plasma** occurs at exceptionally high temperatures and can be found in stars and lightning bolts. It is also present in fluorescent light bulbs and plasma TV. The **Bose-Einstein** condensate is the state that occurs at super low temperatures.

Solids

In solids, the molecules are tightly packed. There is no space between the molecules, so, they cannot move around. Because of this, solids have a fixed shape and a definite volume[1]. However, the shape of some solids can be changed by applying force.

[1]**volume:** the amount of space that an object or substance fills

Examples: Tables, books, and pencils are examples of solids. Certain solids such as sponges, rubber bands, and clay can change their shapes. Sponges and rubber bands regain[2] their original shape when force is removed, but clay does not.



Copper sulphate crystals

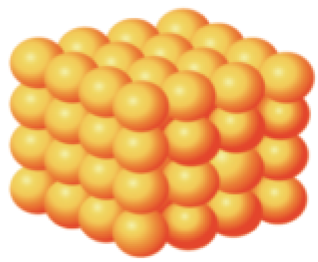
Crystals such as those of **sugar** and **copper sulphate** are also examples of solids. When you observe these crystals using a magnifying lens or under a microscope, you will be able to see their shapes.

Sugar crystals

Properties of Solids

The following are the properties of solids:

- Solids have a fixed shape.
- Solids have a definite volume.
- Solids do not flow.
- The molecules in solids are tightly packed.



Molecular arrangement in solids

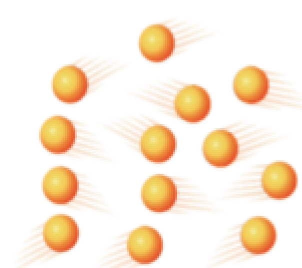
Liquids

In liquids, the molecules are loosely packed. There is some space between the molecules; therefore, they can move around. Thus, liquids do not have a fixed shape and take the shape of the container into which they are poured. Liquids can flow. Liquids have a fixed volume. Milk, water, oil, and petrol are examples of liquids.

Properties of Liquids

The following are the important properties of liquids:

- Liquids do not have a fixed shape. They take the shape of the containers, into which they are poured.
- Liquids have a fixed volume.
- Liquids flow from higher level to lower level.
- The molecules in liquids are loosely packed so they can move around. This property enables liquids to flow.



Molecular arrangement in liquids

[2]**regain:** get back

Activity

Aim: To demonstrate that liquids take the shape of the container

Materials required: Glass containers of different shapes (beakers, measuring cylinders, conical flasks, round flasks, etc.), water, and different water colours

Liquids take the shape of the container

Procedure:

1. Place all the containers on a table.
2. Pour some water into each container.
   Add a drop of colour to each of the containers and mix thoroughly. Use a different colour for each container. Coloured water can be seen more clearly in the container. Observe how the water takes the shape of each container.

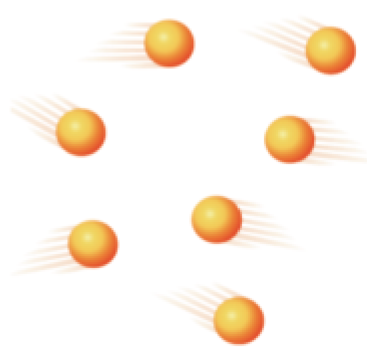
Observation: Water takes the shape of the container in which it is poured.

Conclusion: Liquids do not have a fixed shape and take the shape of the container in which they are poured.

Gases

In gases, the molecules are very loosely packed. There is a lot of space in between the molecules; therefore, they can freely move around. Thus, gases do not have a fixed shape or volume. They can flow easily and fill up the entire space available to them.

Examples: Air and cooking gas are mixtures of gases.



Molecular arrangement in gases

Properties of Gases

The following are the important properties of gases:

- Gases do not have a shape of their own. In an empty container, they occupy all the available space.
- They do not have a fixed volume. They occupy the volume of the container.
- They can flow easily in all directions.
- The molecules in gases are very loosely packed. The presence of a lot of space between the particles enables gases to move around freely.

Think and Discuss

The ability of gases to flow easily can affect how pollution spreads in the air. What environmental impacts could this have?

Questions

Write *T* for True and *F* for False.

1. Copper sulphate crystals are an example of solids. ....
2. In solid state of matter, the molecules are tightly packed. ....
3. Petrol is a liquid state as it does not have a fixed shape, but has a fixed volume. ....
4. The liquid state of matter can fill the entire space available to it. ....
5. Oxygen and nitrogen are solid states of matter. ....

▶ Solution

When two substances are combined together in such a way that they evenly spread and mix with each other, it is called a ***solution****.* Sugar or salt dissolved in water are examples of a solution. Solute and solvent are the two substances that make a solution.

Solute *The substance that gets dissolved in a solution is called the* ***solute****.*

Solvent *The substance that dissolves the solute particles into it to make a solution is called the* ***solvent****.*

Solution	Solute	Solvent
Sugar solution	Sugar	Water
Salt solution	Salt	Water

Types of Solutions

Based on the type of solute and solvent present in a solution, there are different types of solutions. Let us learn about these solutions in the table given below.

Type of solution	Example	Solute	Solvent
Solid in liquid	Sugar water, salt water	Solid (sugar, salt)	Liquid (water)
Gas in liquid	Soda water, soft drink	Gas (carbon dioxide)	Liquid (water)
Gas in gas	Air	Gas (oxygen)	Gas (nitrogen)
Liquid in liquid	Alcohol and water	Liquid (alcohol)	Liquid (water)

Activity

Aim: To make a solution using water as a solvent

Materials required: A glass of water, a spoon of sugar, an empty glass, and a table

Procedure:

1. Place the empty glass on the table.
2. Fill it half with water.
3. Add the sugar into it and stir it with the spoon.

Sugar is the solute and water is the solvent.

Observation: Sugar completely dissolves in water, forming a clear solution without any visible particles remaining.

Conclusion: Water acts as a good solvent in which sugar (solute) is added and a solution is thus formed.

Activity

Aim: To make a solution using turpentine oil as a solvent

Materials required: A disposable paper cup (to be thrown after use), turpentine oil, red paint, a spoon, and a table

Procedure:

1. Under the supervision of an adult, carefully fill one-fourth of the cup with turpentine oil.
2. Place the cup on the table, add a small amount of paint, and stir with a spoon.

Observation: The paint mixes completely with turpentine oil over time. The oil also becomes red in colour. Paint is not seen separate in the oil.

Conclusion: Turpentine oil acts as a good solvent for the red paint (solute). A solution is thus formed.

Activity

Aim: To make a solution using milk as a solvent

Materials required: A paper cup half filled with warm milk, two spoons full of honey, and a table

A solution of honey in milk

Procedure:

1. Place the paper cup containing milk on the table.
2. Pour the honey into it, and stir it with the spoon.

Observation: The honey dissolves completely into the milk after some time. Honey is not seen separate from milk inside the cup.

Conclusion: Milk acts as a good solvent for the honey (solute). A solution is thus formed.

Soluble and Insoluble Substances

Substances that dissolve completely in a liquid to form a solution are called ***soluble substances****.* For example, sugar and salt are soluble in water. *Substances that do not dissolve in a liquid are called* ***insoluble substances****.* For example, sand and chalk are insoluble in water.

Miscible and Immiscible Liquids

When two liquids can be mixed together such that they dissolve completely, they are called ***miscible liquids****.* An example is alcohol and water. *When two liquids are mixed but do not dissolve completely they are called* ***immiscible liquids****.* Examples include water and oil, water and diesel, and water and petrol. The properties of miscible and immiscible liquids can be demonstrated by testing different liquids (vinegar, lemon juice, ink, and mustard oil) with water in test tubes. Vinegar, lemon juice, and ink mix well with water, whereas mustard oil and refined oil do not mix with water and form separate layers.

Separation of Liquids from Solids

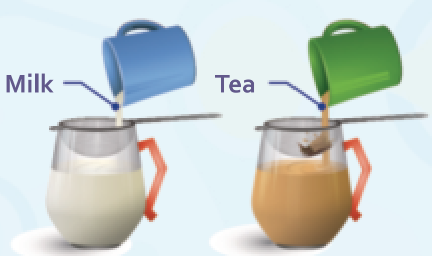
Soluble and insoluble solutes can be separated from solvents using different methods.

Activity

SCIENTIFIC SKILLS · TP · EXPERIENTIAL LEARNING

Aim: To identify whether a substance is soluble or insoluble in water

Materials required: Five cups and a large jug or jar, one plain sheet of paper, stirrer, five strainers, water, tea leaves, milk, common salt, sand, and small twigs



Procedure:

1. Take five cups and fill them with water.
2. Now, add one small teaspoon of common salt to the first cup, tea leaves to the second, milk to the third, sand to the fourth, and twigs to the fifth.
3. Stir the contents in each cup with a stirrer.

4. Now, place the strainer over the jug or jar and strain each of the five cups, one by one. After straining from each cup, transfer the materials collected on the strainer onto a sheet of paper, before starting to strain from another cup.

Observation: Tea leaves, sand, and twigs are left in the strainer, but common salt and milk pass through the strainer and are not left behind.

Conclusion: Tea leaves, sand, and twigs do not dissolve in water and, therefore, are called insoluble substances. Common salt and milk dissolve in water and, therefore, are called soluble substances.

Separation of Soluble Substances

We can remove soluble substances from a liquid (say water) by methods such as boiling and distillation.

Boiling In this process, the solution containing soluble impurities, such as salt and sugar, is heated till the water evaporates, leaving behind salt, sugar, or any other soluble substances. During this process, water is lost through boiling.

SCIENTIFIC SKILLS EXPERIENTIAL LEARNING

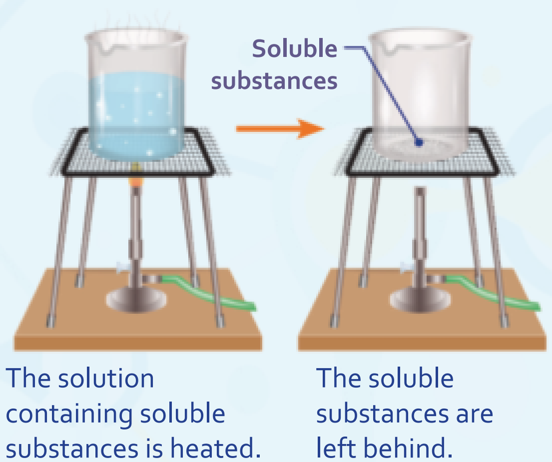
Activity

Aim: To separate soluble substances by boiling

Materials required: A Bunsen burner, a stand, a beaker, wire mesh, about 50 ml of salt solution, and matches

Procedure: [**Note: Adult supervision required**.]

1. Place a stand above the Bunsen burner. Keep a wire mesh over it.
2. Now place the beaker on the wire mesh.
3. Pour salt solution into it.
4. Let your teacher/an adult light the burner using the matchstick.
5. Heat continuously until water evaporates from the beaker.
6. Now, switch off the burner and allow the beaker to cool down. Note what is left behind in the beaker.

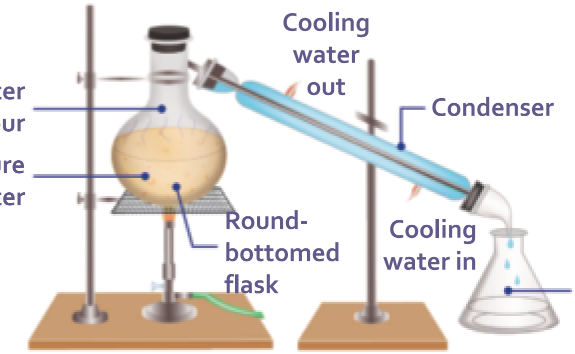


Boiling

Observation: Water has evaporated and salt is left behind in the beaker.

Conclusion: Salt is the solute which is soluble in water (solvent). We have separated salt from its solution through boiling. Thus, boiling can be used to separate soluble impurities from a solution.

Distillation This method is generally performed in laboratories to remove soluble impurities from water. *Pure water collected by the distillation process is called **distilled water**.* It is the purest form of water as it does not contain any impurities. Distilled water is used in laboratories, car batteries, and medicines. However, distilled water is not suitable for drinking as it does not contain minerals that are present in the drinking water.



Distillation

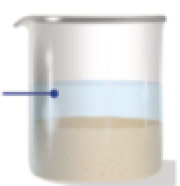
Separation of Insoluble Substances

Insoluble substances, such as mud and sand, can be removed by sedimentation followed by decantation or filtration.

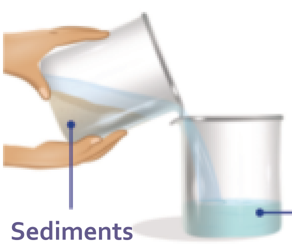
Sedimentation and decantation In sedimentation, insoluble substances in a liquid settle down at the bottom of a container. The substances that settle down are called **sediments**. Sedimentation involves the following steps:

i. Impure water, containing mud, is collected from a pond in a beaker.
ii. The water is allowed to stand undisturbed in the beaker for a while.
iii. *After some time, the mud settles down at the bottom of the beaker as sediment. This process is called **sedimentation**.*

Water above the mud or sand layer is almost clear. This water can then be poured out into a separate container without disturbing the sediments. *The process of removing the clear water after the sediments have settled down is called **decantation**.*



Sedimentation



Decantation

Activity

Aim: To separate insoluble substances from water by sedimentation and decantation

Materials required: A glass of water, one spoon of sand, an empty glass or beaker, and a table

Procedure:

1. Place the glass or beaker containing water on the table.
2. Add the sand and stir it with the spoon.
3. Let it stand for 15 minutes.
4. Without disturbing the sand, gently transfer the water from this glass to an empty glass.

Observation: The sand settles down at the bottom of the glass after 15–20 minutes. The water transferred to another glass is somewhat clear and generally does not contain sand.

Conclusion: Sand is an insoluble impurity in water. It is heavier than water so it settles down at the bottom of the glass. This process is called sedimentation. Then, the clear water is transferred to another glass without disturbing the sediment. This process is called decantation.

Filtration In filtration, the insoluble substances can be removed by passing the solution (say, impure water) through a filter paper. While the insoluble substances are retained on the filter paper, the clear water passes through it and is collected in the flask below.

The process of separating insoluble substances by passing the solution through filter paper is called ***filtration***. This is a better method of removing insoluble substances than sedimentation and decantation.

Filtration

Activity

Aim: To separate insoluble substances by filtration

Materials required: A glass of water, a spoon of sand, an empty glass, a funnel, a beaker, and a filter paper

Procedure:

1. The filter paper is placed inside the funnel as shown in the figure. It is moistened by letting it absorb a few drops of water.
2. Next, sand is added to the glass of water and is stirred with a spoon.
3. This dirty water is then poured on to the funnel.
4. The clear water (filtrate) is collected in the beaker placed just below the funnel.

Observation: The water collected in the beaker is clean.

Conclusion: The filter paper acts as a filter and sand particles get trapped in it. The water comes out clean. This process is called filtration. This process is used to separate larger sized insoluble substances from liquids.

REMEMBERING UNDERSTANDING

Questions

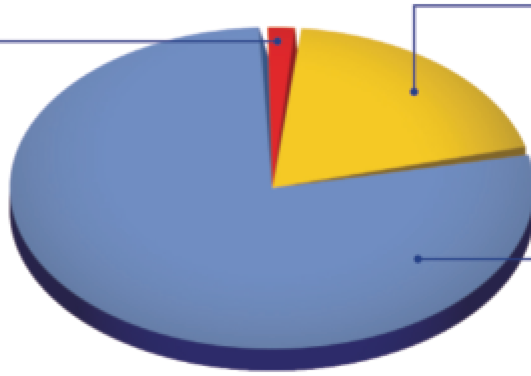
Fill in the blanks.

1. The substance that dissolves completely in a liquid, to form a solution is called ....
2. Tea leaves and sand are .... substances in water.
3. .... is lost through boiling.
4. Distilled water is not suitable for ....
5. The process of separating insoluble substances by passing the solution through filter paper is called. ....

Air

Air is a mixture of several gases—78% of it is nitrogen and 21% of it is oxygen. Rest of it comprises other gases such as carbon dioxide, argon, helium, hydrogen, etc. Water vapour is also present in the air. *The amount of water vapour present in the air at a particular time is called* ***humidity***. Air also contains dust particles and smoke.

Carbon dioxide and other gases A layer of carbon dioxide in the atmosphere acts as a shield preventing heat from leaving Earth. Carbon dioxide is used by green plants to make their own food. Argon is a colourless gas. It is used in making light bulbs and fluorescent tubes. Water vapour, ozone, hydrogen, and helium are some other gases present in the air.



Gases present in the air

Oxygen Oxygen is important for all living things to survive.

Nitrogen Nitrogen is the most abundant gas present in the air. In the absence of nitrogen in the air, even small fires can become uncontrollable. Thus, the presence of nitrogen in the air controls the process of burning. It is also required for the growth of the plants.

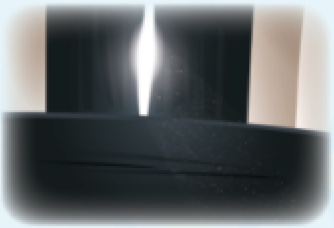
Activity

Aim: To show the presence of dust particles in air

Materials required: A room and a torch

Procedure:

1. Go inside the room and close all the doors and windows.
2. Switch off all the lights and fans in the room. Close all the curtains and make the room dark enough.
3. Switch on the torch light and see if anything can be seen in the beam of torch light.



Tyndall effect with dust particles visible in light

Observation: Small dust particles are seen moving around in the beam of torch light.

Conclusion: Air contains dust particles.

Activity

Aim: To show the presence of water vapour in air

Materials required: A glass, few ice cubes, and a table

Procedure:

1. Place the glass on the table and pour ice cubes into it.
2. Wait for 5–10 minutes and then observe the glass.

Observation: Small water droplets are seen on the outer surface of the glass.

Conclusion: Outer surface of the glass turns cool due to ice inside it. So, water vapour that is generally present in the air gets condensed on the cool outer surface. This proves air contains water vapour.

Properties of Air

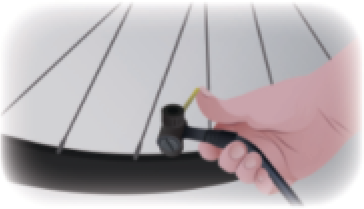
The following are the properties of air.

- Air occupies space.
- Air exerts pressure.
- Air is needed for burning.
- Air has weight.

Fact File

Anemometer measures the wind speed.

Three balloons containing varying amounts of air: less, more, and maximum



Air filled in a bicycle tyre

Air filled in a truck tyre

The activity given below demonstrates that air occupies space.

Activity

Aim: To show that air occupies space

Materials required: A bucket, water, and a glass

Procedure:

1. Fill the bucket with water.
2. Now hold the glass in an upside-down position over the bucket.
3. Push the glass straight into the bucket without tilting it.
4. Observe whether water enters into the glass.
5. Now tilt the glass slightly and observe.

Air occupies space.

Observation: During the first step, water does not enter the glass, no matter how much it is pushed into the water. But when the glass is tilted, bubbles of air escape from the glass and water fills the space inside the glass.

Conclusion: This shows that air occupies space.

Air Rises on Heating

Another important property of air is that warm air is lighter than cold air. Because of this, when air gets heated, it rises up. This property leads to the formation of wind.

Formation of Wind

Sun's heat causes the air to become warm. As the warmer air is lighter, it starts rising up. The cooler air, being heavier than the warmer air, starts moving in to take its place. This causes wind.

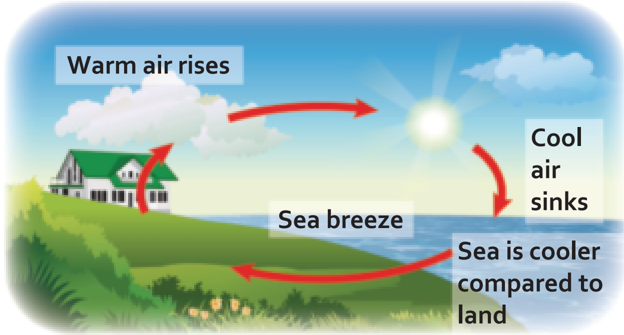
- *Moving air is called **wind**.*
- *A gentle wind is called a **breeze**.*
- *A strong wind is called a **gale**.*
- *A very strong and powerful wind with rain is called a **storm**.*
- *A storm that occurs with thunder and lightning is called a **thunderstorm**.* Wind affects the weather of a place.

Fact File

Weather vanes are used to indicate the wind direction. It is also called wind vane, or weathercock.

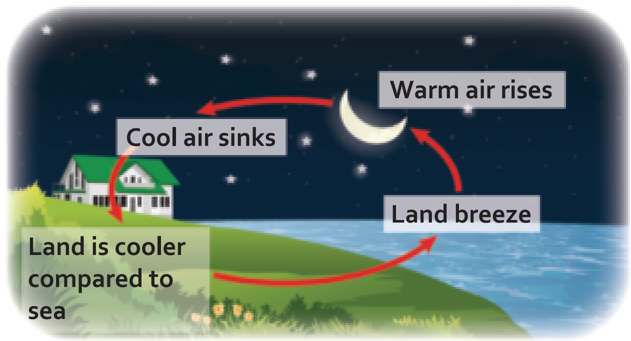
Sea and Land Breezes

During the day, the land heats up faster than the sea, and so does the air above the land. Since warm air is lighter, the air above the land rises. The cooler and heavier air above the sea rushes over the land to fill up the space. *The cooler air that moves from the sea towards the land during daytime is called* ***sea breeze****.*



Sea breeze

During the night, the land cools down faster than the sea. Since the sea is warmer than the land at night, the air above the sea also remains warm. The warm air is lighter and rises. The cooler and heavier air above the land moves towards the sea to fill up the space. *The cooler air that moves from the land towards the sea during the night is called* ***land breeze****.*

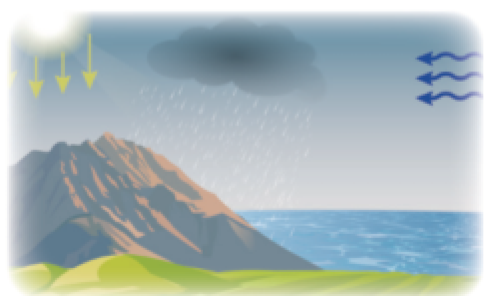


Land breeze

Monsoon Breeze (Monsoon Wind)

KNOWLEDGE OF INDIA

In places near the equator, during summer the land warms up faster than the sea or ocean due to direct sun rays. Since the temperature of the land is more than the sea or the ocean, the air above it also gets heated up. We know that warm air rises up, so the warm air above the land rises up. The moisture-laden cold wind from the sea moves towards the land to take the place of the risen warm air. *The moisture-laden air that moves from the sea towards the land during summers is called* ***monsoon breeze*** *or* ***monsoon wind****.* These monsoon winds bring rain. In India, we get rainfall during the rainy season due to these winds. During the winter season, the reverse happens. The land cools faster than the sea. So the cool, heavy, and



Summer monsoon wind

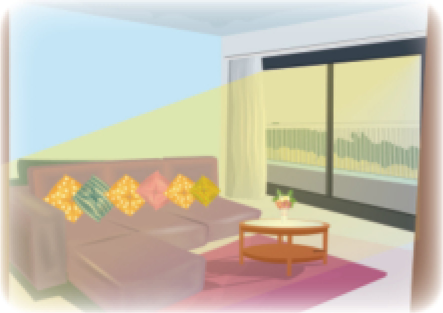
Winter monsoon wind

Monsoon breeze

dry air above the land travels towards the sea, while the hot air above the sea rises up. So, the air that blows over the land in the winter is dry monsoon wind. Hence, no rainfall occurs because of these winds.

Role of Ventilators in Closed Spaces

Ventilators and exhaust fans are generally located at the top portion of the walls of rooms. This is because the air that we breathe out is warmer and impure as it is rich in carbon dioxide. So, it rises up and goes out through the ventilators. Fresh air from outside then enters the room though windows. Thus, we should keep our windows open and let fresh air and sunlight come in.



A well-ventilated house

Air pollutants, unwanted smells of cooking, and foul smell also escape through ventilators. If you do not have adequate air flow in your home, then moisture will collect and dampen your walls, wooden furniture, and other essential items of your home. Moulds can also grow and cause a health hazard. Thus, to keep our homes fresh, dry, and airy we should have ventilators and exhaust fans in our homes.

In the kitchens in our homes and also in factories, chimneys and exhaust fans are used to assist in throwing out smoke and foul air out, so that fresh air can enter from outside and take its place.

Questions

REMEMBERING UNDERSTANDING

Write *T* for True and *F* for False.

1. Air is a mixture of several gases such as nitrogen, oxygen, carbon dioxide, etc. ....
2. Air contains dust particles and smoke. ....
3. Oxygen is taken in by green plants to prepare food. ....
4. Nitrogen in air controls the process of burning. ....
5. Air has no weight. ....
6. A gentle wind is called gale. ....
7. The land breeze takes place during daytime. ....
8. Summer monsoon wind brings rain in India. ....

Wrap Up

- Anything that has mass and occupies space is called matter. Solid, liquid, and gas are the three states of matter.
- In solids, molecules are tightly packed, giving them a fixed shape and definite volume. In liquids, molecules are loosely packed. Liquids have a fixed volume but no fixed shape and can flow. In gases, molecules are very loosely packed. Gases have no fixed shape or volume and fill any available space.
- When two substances are combined together in such a way that they evenly spread and mix with each other, it is called a solution.
- Substances that dissolve completely in a liquid to form a solution are called soluble substances. Substances that do not dissolve in a liquid are called insoluble substances.
- Soluble substances can be removed from a liquid by boiling or distillation. Insoluble substances can be removed by sedimentation followed by decantation or filtration.
- Air is a mixture of several gases.
- Sea breeze is the cooler air that moves from the sea to the land during the day. Land breeze is the cooler air that moves from the land to the sea during the night. Monsoon breeze is the moisture-laden air that moves from the sea to the land during summers.
- The air we breathe out is warmer and rich in carbon dioxide, rising and exiting through ventilators at the top of a room.
- In kitchens and factories, chimneys and exhaust fans help remove smoke and foul air, allowing fresh air to enter through windows.

Exercises

SECTION I

REMEMBERING UNDERSTANDING APPLYING

A Choose the correct option to fill in the blank.

1. Matter is made of molecules that are very .... (large/small).
2. In solids, there is .... (no/some) space between the molecules.
3. Copper sulphate crystals are .... (solids/liquids) .
4. Solids and................ (gases/liquids) have fixed volume.
5. Solute and .... (gas/solvent) makes a solution.

B Assertion and Reasoning questions

1. **Assertion (A):** Filtration can be used to separate a mixture of sand and water.

   Reason (R): Sand particles are smaller than water molecules and can pass through the filter paper.

   a. Both A and R are True b. Both A and R are False
   c. A is True and R is False d. A is False and R is True

2. **Assertion (A):** Ventilators help maintain good air quality in closed spaces.

   Reason (R): They only remove warm air and replace it with cooler air.

   a. Both A and R are True b. Both A and R are False
   c. A is True and R is False d. A is False and R is True

C Match the following.

1. Gas	i. Soluble in water
2. Solution	ii. Obtained by distillation process
3. Chalk	iii. Flows in all direction
4. Pure water	iv. Insoluble in water
5. Honey	v. Sugar or salt dissolved in water

SECTION II

REMEMBERING UNDERSTANDING APPLYING

D Short answer questions

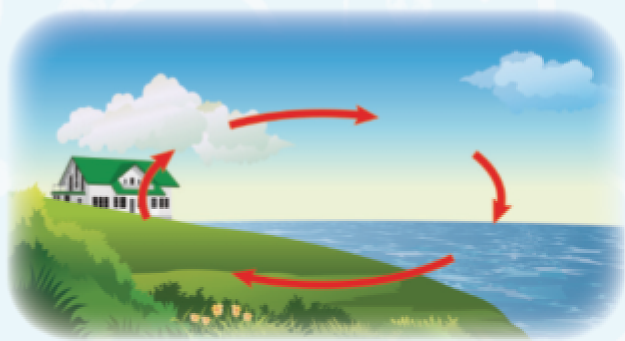
1. Is the arrangement of molecules in all states of matter the same?
2. What is a solution? Give two examples.
3. Shubham is making a delicious cup of hot chocolate. The cocoa powder mixes smoothly with the hot milk, but some leftover marshmallows just float on top. What kind of substances can the cocoa powder and marshmallows be classified as here?
4. Neha has a bucket of rainwater that collected some dirt and leaves. How could she remove the dirt and leaves so she can use the water to clean the terrace floor? What is this process called?
5. What are the different properties of air?

E Long answer questions

1. Write three properties each of solids, liquids, and gases. Can you think of everyday objects that demonstrate these properties?
2. Write an activity to identify soluble and insoluble substances in water.
3. Explain two methods to separate soluble substances.
4. How are sea breeze and land breeze formed?
5. What are monsoon winds? How are they helpful to us?

Picture Study

Look at the picture given here and answer the following questions.



1. Does the picture depict sea breeze or land breeze?

   ....

2. When does it occur—day or night? Draw a sun or moon accordingly.

   ....

3. Explain briefly what is happening in the picture.

   ....

My Learning Corner

A Think about

The science teacher asked the students, "When you spray perfume, it spreads in all directions. Can you explain why?"

....

B Try out

1. Draw any ten pictures of various household items in your Science scrapbook and classify them into solids, liquids, and gases. Also, use bindis to show the molecule arrangements of each category (solids, liquids, and gases).

2. Take a balloon and blow air into it. Observe the change (**less, medium, maximum**) in the volume of balloon at three different stages (as more and more air enters it!).

   Complete the following table based on your observation by using suitable terms given above in bold. Perform the activity above, and write down the observation in your Science scrapbook.

Stage	Amount of air in balloon	Change in volume
I	Less air, small size of the balloon	.... volume
II	More air, medium size of the balloon	.... volume
III	Maximum air, largest size of the balloon	.... volume

3. In your notebook, write four examples of solutions that we use in our day-to-day life. Also, think and write the solute and solvent present in it.

Self-Assessment

Now that you have completed the chapter, score each of the following tasks from 1 to 5 to indicate how well you can do them.

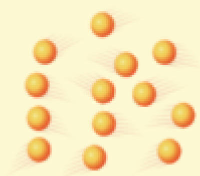
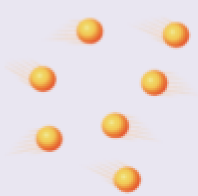
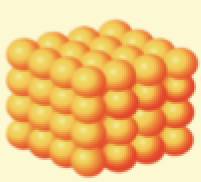
Score 5 = I can definitely do this.

Score 1 = I cannot do this yet.

I can...	My score
• state simple properties of solids, liquids, and gases and demonstrate them through simple activities.	
• identify the different forms of matter and provide examples of each based on observable properties.	
• describe the composition of air and depict it diagrammatically.	
• give examples of warm and fresh air in daily life.	
• differentiate between wind, breezes, and storms with examples.	
• explain the need for ventilators and windows in buildings.	
• relate the use of fans, ACs, and coolers in seasons.	

Worksheet

Observe carefully the pictures of molecular arrangement given below. Identify the state and give three examples and three important characteristics for each state.

Molecule arrangement	Identify the state	Three examples	Any three important features
	....	....	....
	....	....	....
	....	....	....